2 marks

1. What is Human-Centric Design Thinking?

- Human-centric design thinking is an approach that focuses on understanding the **needs, feelings, and problems of users**.
- It develops solutions based on **real user experiences**.
- It involves empathy, problem identification, idea generation, prototyping, and testing.
- The main goal is to create **useful, simple, and user-friendly solutions**.

2. Give an example of a case study that demonstrates Human-Centric Design principles.

Example: ATM Design

- Designers observed that users found ATM operations confusing.
- They studied users' difficulties and simplified the screen instructions.
- Large buttons, clear messages, and simple steps were introduced.
- This made ATM usage **easier, faster, and more accessible** for users.

3. Give any real-life example for Empathy.

Example: Helping an Elderly Person

- An elderly person may find it difficult to use a smartphone.
- Understanding their difficulty is an example of **empathy**.
- A simple interface with large icons and clear instructions can be designed.
- This solution considers the **user's feelings and needs**.

4. What are the stages of the Design Thinking process?

The main stages are:

1. **Empathize** – Understand the users and their needs.
2. **Define** – Clearly identify the problem.
3. **Ideate** – Generate different possible solutions.
4. **Prototype** – Create a simple model of the solution.
5. **Test** – Test the prototype with users and improve it.

5. Importance of Customer-Centric Innovation

- It helps businesses understand **customer needs and expectations**.

- It improves **customer satisfaction and experience**.
- It helps develop products that solve **real customer problems**.
- It increases **customer loyalty and business success**.

6. Define Problem Validation with Example

- **Problem validation** is the process of checking whether an identified problem is **real, important, and experienced by customers**.
- It is done through customer interviews, surveys, and observation.
- It prevents businesses from solving problems that do not actually exist.
- **Example:** Interviewing students to confirm whether they really face difficulties in finding college bus timings before developing a bus-tracking app.

7. Problem Incidence with Some Examples

- **Problem incidence** refers to how frequently a particular problem occurs among users.
- It helps determine whether a problem is common or occurs only occasionally.
- A frequently occurring problem may have greater importance for innovation.
- **Examples:** Long queues at hospitals, traffic congestion, difficulty in online payment, and delays in food delivery.

8. Customer Interviews and Field Visit

- **Customer interviews** involve directly asking customers about their needs, problems, and experiences.
- They help collect detailed information from the actual users.
- **Field visits** involve observing customers in their real working or living environment.
- Both methods help designers understand **real-world problems and user behaviour**.

9. Explain the Concept of Minimum Usable Prototype (MUP)

- A **Minimum Usable Prototype (MUP)** is the simplest version of a product that contains the **essential features needed for users to try it**.
- It is created with minimum time and resources.
- Users can test the prototype and provide feedback.
- The feedback is used to **improve the final product**.

Example: A food delivery app prototype may initially provide only restaurant selection, food ordering, and order tracking.

10. Techniques of Value Proposition Design

- **Customer Profile:** Identify the customer’s needs, problems, and expectations.
- **Value Map:** Identify products or services that provide value to customers.
- **Pain Relievers:** Explain how the product reduces customer problems.
- **Gain Creators:** Explain how the product provides additional benefits and satisfaction.

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1. Define Human-Centered Design (HCD).

- **Human-Centered Design (HCD)** is a problem-solving approach that focuses on the **needs, feelings, experiences, and problems of users**.
- It involves users throughout the design process to understand their actual requirements.
- It includes stages such as **empathise, define, ideate, prototype, and test**.
- The main aim of HCD is to create solutions that are **useful, usable, and satisfying** for people.

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2. What is the Forge Innovation Rubric (FIR)?

- **Forge Innovation Rubric (FIR)** is a framework used to **assess and diagnose risks in an innovation idea**.
- It helps determine whether the identified customer problem is real and significant.
- It considers factors such as **problem validation, problem significance, customer motivation, value proposition, and problem–solution fit**.
- It helps entrepreneurs improve their ideas before investing significant resources.

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3. Differentiate Empathy and Sympathy.

Empathy	**Sympathy**
Empathy means **understanding another person's feelings and experiences** from their point of view.	Sympathy means **feeling concern or pity for another person's situation**.
It involves trying to understand what the person is experiencing.	It involves expressing concern about what happened to the person.

Empathy	Sympathy
Empathy is important for understanding users in design thinking.	Sympathy may show care but does not necessarily provide deep understanding.
Example: “I understand how difficult this problem is for you.”	Example: “I am sorry that you are facing this problem.”

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4. State any two interviewing techniques used in Design Thinking.

1. Open-Ended Questions

- Open-ended questions allow users to explain their thoughts, experiences, and problems in their own words.
- Example: **“Can you explain the difficulties you face while using this service?”**

2. Probing Questions

- Probing questions are follow-up questions used to obtain more detailed information.
- Example: **“Why was that experience difficult for you?”**

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5. What is Customer Validation?

- **Customer Validation** is the process of checking whether customers **accept, need, and are willing to use or purchase a proposed product or solution**.
- It is usually performed after understanding and validating the customer problem.
- It uses methods such as **customer interviews, surveys, prototype testing, and feedback**.
- The main purpose is to confirm that the proposed solution provides real value to customers.

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6. Differentiate Customer Discovery and Customer Validation.

Customer Discovery	Customer Validation
It focuses on **understanding customers and identifying their problems and needs**.	It focuses on **testing whether the proposed solution satisfies customers**.

Customer Discovery	Customer Validation
It answers **“Who are the customers and what problems do they have?”**	It answers **“Will customers accept and use the solution?”**
Interviews, observation, and surveys are commonly used.	Prototype testing, feedback, and early product trials are commonly used.
It usually occurs **before solution validation**.	It follows customer and problem discovery and tests the proposed solution.

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7. Define Problem Validation.

- **Problem Validation** is the process of checking whether an identified customer problem is **real, significant, and experienced by the target users**.
- It ensures that the problem is not simply based on assumptions.
- Interviews, surveys, observations, and field research can be used for validation.
- The aim is to confirm that solving the problem would provide **real value to customers**.

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8. What is a Field Visit? Give a real-time example.

- A **Field Visit** is a research method in which designers visit the **actual place where users perform their activities**.
- It helps designers directly observe user behaviour, problems, and environmental conditions.
- It provides real-world information that may not be obtained through interviews alone.
- **Example:** A team designing a college canteen ordering system visits the canteen during lunch time to observe queues, ordering difficulties, payment problems, and waiting time.

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9. Define Minimum Usable Prototype (MUP).

- A **Minimum Usable Prototype (MUP)** is the simplest version of a product that contains enough essential features to be **usable and tested by real users**.
- It focuses on solving the user's main problem with minimum features.
- It allows designers to collect early user feedback and identify improvements.

- **Example:** An online food-ordering MUP may provide restaurant selection, food ordering, basic payment, and order tracking.

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10. Why is Value Proposition Important?

- A **value proposition** clearly explains the **main benefit that a product or service provides to its customers**.
- It helps customers understand why they should choose the product over other alternatives.
- It connects the product's benefits with the customer's important needs or problems.
- A strong value proposition helps improve **customer interest, product acceptance, and competitive advantage**.

Human-Centered Design (HCD)

1. What is Customer-Centered Design?

Customer-Centered Design is an approach to designing products, services, or solutions by focusing mainly on the **needs, wants, expectations, and problems of customers**.

- It gives importance to understanding the customer before developing a solution.
- Customer feedback is collected and used throughout the design process.
- The main aim is to provide a **useful, convenient, and satisfying customer experience**.
- Example: A food-delivery app studies customer problems such as difficult ordering, late delivery, and complicated payment, and designs features to solve them.

2. What is Human-Centered Design?

Human-Centered Design (HCD) is a problem-solving approach that focuses on the **needs, experiences, abilities, and challenges of people**.

- It involves users throughout the design process.
- It tries to understand the real problems faced by people.
- Solutions are developed based on user needs rather than assumptions.
- The solution is tested with users and improved based on their feedback.

Example: When designing an ATM, designers consider the user's need for **simple navigation, clear instructions, accessibility, security, and quick transactions**.

3. Human-Centered Design Process

The HCD process generally consists of **five important steps**:

1. Empathise

- Empathising means understanding the user's **feelings, needs, problems, and experiences**.
- Designers observe users, conduct interviews, and collect information about their problems.
- The aim is to understand the problem from the user's point of view.

2. Define

- In this stage, the information collected from users is analysed.
- The main problem is clearly identified and defined.
- A clear **problem statement** is prepared.
- Example: “Students need an easier way to find their classroom quickly.”

3. Ideate

- Ideation means generating different ideas to solve the identified problem.
- Brainstorming and creative thinking techniques are used.
- Many possible solutions are considered before selecting the best idea.

4. Prototype

- A prototype is a **simple model or early version of the proposed solution**.
- It helps designers understand how the solution will work.
- Prototypes can be sketches, paper models, wireframes, or working models.

5. Test and Iterate

- The prototype is given to users for testing.
- Users provide feedback about what works and what does not work.
- Designers make changes based on the feedback.
- This process is repeated until a better solution is developed.

4. Three Main Phases of Human-Centered Design

Human-Centered Design can also be divided into **three major phases**:

1. Inspiration Phase

- The inspiration phase focuses on **understanding the users and their problems**.
- Designers interact with users through interviews, observation, and research.
- It helps designers discover real user needs and opportunities.

2. Ideation Phase

- The ideation phase focuses on **generating and developing possible solutions**.
- Different ideas are discussed and evaluated.
- Prototypes are created to represent the selected ideas.

3. Implementation Phase

- The implementation phase focuses on **putting the final solution into practice**.
- The solution is tested with real users.
- Feedback is collected and improvements are made before final implementation.

5. Principles of Human-Centered Design

1. Understand the Users

Designers should clearly understand the needs, problems, and expectations of users.

2. Focus on User Needs

The design should be based on actual user requirements rather than assumptions.

3. Involve Users

Users should participate in the design process by providing opinions and feedback.

4. Identify the Real Problem

Designers should find the actual problem before developing a solution.

5. Generate Multiple Ideas

Different possible solutions should be considered before selecting the best one.

6. Prototype Early

Simple prototypes should be created early to identify problems quickly.

7. Test with Users

The proposed solution should be tested with real users to understand its effectiveness.

8. Improve Continuously

Design should be an iterative process where feedback is used to continuously improve the solution.

9. Make the Design Simple

The solution should be easy to understand, learn, and use.

10. Consider Accessibility

The design should be usable by people with different abilities and requirements.

6. Examples of Human-Centered Design

Example 1: ATM

ATM interfaces are designed with simple menus, clear instructions, and easy-to-understand buttons to help users complete transactions easily.

Example 2: Mobile Banking App

Mobile banking applications provide simple navigation, quick payment options, biometric login, and transaction history based on user requirements.

Example 3: Food Delivery App

Food delivery applications allow users to search restaurants, view menus, track orders, and make payments conveniently.

Example 4: Wheelchair Ramp

Wheelchair ramps are designed after considering the mobility needs of people who use wheelchairs.

Example 5: Online Learning Platform

Learning platforms provide simple navigation, recorded classes, quizzes, and progress tracking to support students' learning needs.

7. Conclusion

Human-Centered Design is a user-focused and iterative approach to problem solving. It starts by understanding people, defines their problems, generates ideas, develops prototypes, and tests them with users. The main goal is to create solutions that are **useful, usable, accessible, and satisfying for people**.

Innovation Risk Diagnostic – Forge Innovation Rubric (FIR)

Introduction

Forge Innovation Rubric (FIR) is an innovation risk diagnostic tool used to evaluate whether an idea or innovation has a strong chance of becoming successful. It helps identify **risks, weaknesses, and opportunities** in an early-stage business idea.

The FIR mainly examines whether the **problem is real, important to customers, and properly matched with the proposed solution**.

Example Used:

Online Food Delivery App for College Students

1. Problem Validation

- Problem validation means checking whether the **identified problem actually exists** for the target customers.
- The entrepreneur collects information through **interviews, surveys, observation, and customer feedback**.
- A problem should not be based only on assumptions.
- The problem should occur frequently enough to require a solution.
- **Example:** College students are asked about difficulties in getting food during short breaks, and many students say that waiting in long queues is a common problem.

2. Problem Significance

- Problem significance measures **how important and serious the problem is to customers**.
- A significant problem usually causes inconvenience, loss of time, extra cost, or dissatisfaction.
- Customers are more likely to accept a solution when the problem has a strong impact on their daily life.
- The entrepreneur should identify the consequences of not solving the problem.
- **Example:** Students may lose their limited break time by standing in long food queues, making the problem significant for them.

3. Customer Motivation

- Customer motivation explains **why customers would want to solve the identified problem**.
- Customers should have a clear reason to use or purchase the proposed solution.
- Motivation may come from saving **time, money, effort, or improving convenience**.
- Strong customer motivation increases the possibility of adoption.

- **Example:** Students are motivated to use the food delivery app because they can order food before their break and avoid waiting in a queue.

4. Value Proposition

- A **value proposition** explains the main benefit that the solution provides to customers.
- It clearly describes **why customers should choose the proposed solution**.
- A good value proposition should be simple, useful, and different from existing alternatives.
- It should directly connect the solution with the customer's important need.
- **Example:** “Get your food delivered to your college location during your break, saving time and avoiding long queues.”

5. Problem–Solution Fit

- Problem–solution fit means checking whether the **proposed solution effectively solves the identified customer problem**.
- The solution should directly address the main problem rather than solving an unrelated issue.
- Customer feedback and prototype testing are used to check the effectiveness of the solution.
- If the solution does not satisfy customers, it should be modified and tested again.
- **Example:** The food delivery app provides pre-ordering and scheduled delivery, which directly solves the problem of students waiting in long queues.

FIR Flow

Identify Problem → Validate Problem → Check Problem Significance → Understand Customer Motivation → Define Value Proposition → Check Problem–Solution Fit → Test and Improve

Conclusion

The **Forge Innovation Rubric (FIR)** helps entrepreneurs identify innovation risks before investing significant time and resources. By checking **problem validation, problem significance, customer motivation, value proposition, and problem–solution fit**, an entrepreneur can determine whether an innovation is solving a **real and important customer problem**.

Empathy Building Techniques and Strategies

Introduction

Empathy building is the process of understanding the **needs, feelings, thoughts, problems, and experiences of customers or users**. It helps designers look at a problem from the customer's point of view and develop better solutions.

1. Empathy Maps

- An **empathy map** is a visual tool used to understand what a customer **says, thinks, does, and feels**.
- It helps designers understand the customer's emotions, expectations, and difficulties.
- Information is collected through interviews, observations, and customer feedback.
- It helps identify hidden needs that customers may not directly express.
- **Example:** For a food delivery app, the map may show that students say “delivery is slow,” think “I may not get food during my break,” do “check delivery status repeatedly,” and feel “frustrated.”

2. Customer Journey Mapping

- **Customer journey mapping** represents the different stages a customer goes through while using a product or service.
- It identifies customer actions, feelings, problems, and interactions at each stage.
- It helps designers find **pain points and opportunities for improvement**.
- The journey can include stages such as awareness, selection, purchase, usage, and feedback.
- **Example:** For an online shopping service, the journey may include searching for a product, selecting the product, placing an order, receiving the product, and giving feedback.

3. Mind Maps

- A **mind map** is a visual technique used to organize thoughts and information around a central topic.
- The main problem is placed at the centre, and related ideas are connected using branches.
- It helps designers explore different aspects of a customer problem.
- It supports creative thinking and helps identify relationships between different problems and needs.

- **Example:** For “student transportation,” branches can include cost, safety, travel time, availability, comfort, and convenience.

4. Experience Map

- An **experience map** shows how a user experiences a product, service, or situation over a period of time.
- It focuses on the user's **actions, emotions, expectations, and difficulties**.
- It helps designers understand the complete experience rather than focusing on only one interaction.
- It can reveal opportunities for improving the overall customer experience.
- **Example:** In a hospital experience map, the stages may include booking an appointment, travelling to the hospital, registration, consultation, payment, and receiving medicines.

5. Service Blueprint

- A **service blueprint** is a detailed visual representation of how a service is delivered to the customer.
- It shows both the **customer-facing activities and the internal activities** that support the service.
- It helps identify gaps, delays, and problems in service delivery.
- It connects customer actions with employee actions, technology, and supporting processes.
- **Example:** In food delivery, the blueprint can show customer ordering, restaurant preparation, delivery partner pickup, delivery, payment, and customer feedback.

Empathy Building Strategies

1. Conduct User Interviews

- Designers should talk directly with customers to understand their needs, opinions, and problems.
- Open-ended questions should be used to encourage customers to explain their experiences.

2. Observe Users

- Designers should observe customers while they perform real activities.
- Observation helps identify problems that customers may forget or fail to explain during interviews.

3. Collect Customer Feedback

- Feedback should be collected through surveys, reviews, interviews, and discussions.
- The feedback helps designers understand customer satisfaction and pain points.

4. Use Real-Life Experiences

- Designers should experience the product or service from the customer's perspective whenever possible.
- This helps them understand the difficulties faced by users.

5. Identify Pain Points

- Designers should identify situations where customers experience confusion, delay, inconvenience, or dissatisfaction.
- These pain points can become opportunities for innovation.

6. Continuously Improve

- Empathy should be developed continuously throughout the design process.
- New customer feedback should be used to improve the product or service.

Conclusion

Empathy building techniques such as **empathy maps, customer journey maps, mind maps, experience maps, and service blueprints** help designers understand customers deeply. By using strategies such as **interviews, observation, feedback collection, and identifying pain points**, designers can create solutions that are more **useful, convenient, and customer-focused**.

Customer-Centric Innovation (CCI)

1. What is Customer-Centric Innovation?

Customer-Centric Innovation (CCI) is an approach to innovation that focuses on **understanding customer needs, problems, expectations, and experiences** before developing a product or service.

- CCI gives the highest importance to the **customer's needs and satisfaction**.
- It involves customers in different stages of the innovation process.
- Customer feedback is used to improve existing products or create new solutions.
- The main aim of CCI is to create **value for customers and achieve business success**.

- It helps organizations develop products that solve **real customer problems**.

Example: A food delivery company studies students' problems with long waiting times and introduces a pre-order and scheduled-delivery feature.

2. Importance of Customer-Centric Innovation

1. Understands Customer Needs

- CCI helps organizations understand what customers actually need and expect.
- It reduces the chance of developing products based only on assumptions.

2. Solves Real Problems

- It focuses on identifying and solving genuine customer problems.
- This makes the innovation more useful and relevant.

3. Improves Customer Satisfaction

- Products and services designed according to customer needs provide a better experience.
- Satisfied customers are more likely to continue using the product.

4. Reduces Innovation Risk

- Customer feedback helps identify problems at an early stage.
- This reduces the risk of investing in products that customers do not want.

5. Creates Competitive Advantage

- Companies can differentiate themselves by providing better solutions and experiences.
- This helps them compete effectively in the market.

6. Supports Continuous Improvement

- Regular customer feedback helps organizations continuously improve their products and services.

3. Benefits of Customer-Centric Innovation

1. Better Product Quality

- Customer feedback helps organizations identify and correct product weaknesses.
- This results in products that better match customer expectations.

2. Increased Customer Loyalty

- Customers feel valued when companies listen to their opinions.
- This can increase customer trust and loyalty.

3. Higher Market Acceptance

- Products developed according to customer needs have a better chance of being accepted by the market.

4. Increased Business Growth

- Satisfied customers can lead to repeat purchases and increased demand.
- This can contribute to long-term business growth.

5. Efficient Use of Resources

- Customer research helps organizations focus resources on features and solutions that provide real value.

6. Encourages Innovation

- Customer problems and feedback can provide new ideas for products, services, and features.

4. Customer-Centric Innovation in Action

CCI in action means applying customer-centric principles in a real business or product development situation.

Example: Online Food Delivery App

Step 1 – Understand the Customer

- The company interviews college students and finds that students have difficulty ordering food during short breaks.

Step 2 – Identify the Problem

- The main problem is identified as **long waiting time and difficulty getting food during college breaks**.

Step 3 – Generate Ideas

- The company generates ideas such as pre-ordering, scheduled delivery, and live order tracking.

Step 4 – Develop a Prototype

- A simple version of the pre-ordering feature is developed and given to selected students.

Step 5 – Collect Feedback

- Students test the feature and provide feedback about ordering, delivery time, and usability.

Step 6 – Improve the Solution

- The company modifies the feature based on customer feedback.

Step 7 – Launch the Solution

- The improved feature is introduced to a larger group of customers.

Step 8 – Continuous Improvement

- Customer reviews and usage data are continuously analysed to improve the service.

CCI Process

Understand Customers → Identify Problems → Generate Ideas → Develop Solution → Test with Customers → Collect Feedback → Improve → Launch

Conclusion

Customer-Centric Innovation places the customer at the centre of the innovation process. It helps organizations **understand real customer problems, reduce innovation risks, improve customer satisfaction, and create valuable products and services**. Therefore, CCI is an important approach for achieving both **customer satisfaction and long-term business success**.

Customer Discovery (CD)

1. What is Customer Discovery?

Customer Discovery (CD) is the process of understanding **customers, their needs, problems, expectations, and buying behaviour** before developing or launching a product or service.

- It helps entrepreneurs find out whether their business idea solves a **real customer problem**.
- Customer Discovery involves directly interacting with potential customers.
- Interviews, surveys, observations, and feedback are commonly used to collect customer information.
- The main aim is to **validate the problem, understand the customer, and reduce business risk**.

Example:
A startup wants to develop a food-ordering app for college students. Before developing the

app, the team talks to students to understand their problems with food ordering, delivery time, price, and payment.

2. Benefits of Customer Discovery

1. Understands Customer Needs

- CD helps entrepreneurs understand the actual needs and expectations of customers.

2. Identifies Real Problems

- It helps identify whether the problem being addressed is genuinely faced by customers.

3. Reduces Business Risk

- Early customer feedback reduces the risk of developing an unwanted product.

4. Improves Product Design

- Customer opinions help entrepreneurs develop features that provide greater value.

5. Identifies Target Customers

- CD helps identify the specific group of customers who are most likely to use the product.

6. Saves Time and Money

- Testing an idea early prevents unnecessary investment in unsuitable products.

7. Finds New Opportunities

- Customer conversations may reveal additional problems and new business opportunities.

sss3. Customer Discovery Process

Step 1 – Identify the Target Customer

- First, the entrepreneur identifies the group of people who may use the product or service.

Step 2 – Identify Customer Problems

- The entrepreneur identifies the problems, needs, and difficulties faced by the target customers.

Step 3 – Conduct Customer Research

- Interviews, surveys, observations, and discussions are conducted to collect information.

Step 4 – Validate the Problem

- The collected information is analysed to check whether the problem is real and important.

Step 5 – Develop a Solution

- A possible solution is developed based on the validated customer problem.

Step 6 – Test the Solution

- A prototype or minimum viable product (MVP) is tested with potential customers.

Step 7 – Collect Feedback

- Customer feedback is collected to understand what works and what needs improvement.

Step 8 – Improve or Change the Idea

- The product or business idea is improved based on customer feedback.

4. Types of Customer Discovery Approaches

1. Customer Interviews

- Entrepreneurs directly talk with customers to understand their problems, needs, and experiences.

2. Surveys and Questionnaires

- Surveys are used to collect information from a larger number of customers.

3. Observation

- Customers are observed while performing real activities to understand their actual behaviour.

4. Focus Groups

- A small group of potential customers discusses their opinions about a product, problem, or idea.

5. Online Research

- Online reviews, social media comments, forums, and search trends can provide useful customer information.

6. Prototype Testing

- A prototype is given to potential customers to observe how they use it and collect their feedback.

5. Analysing Data and Identifying Patterns

- After collecting customer information, the data should be **organized and analysed**.
- Similar responses and repeated problems should be grouped together.
- Common customer needs, complaints, preferences, and behaviours should be identified.
- Positive and negative feedback should be studied separately.
- Important patterns can reveal the most common and significant customer problems.
- The entrepreneur should focus on problems that occur frequently and have a strong impact on customers.

Example:
If many students say that food delivery is expensive, slow, and difficult to track, these repeated responses indicate important customer needs.

6. Applying Customer Discovery Insights to Your Startup

Customer Discovery insights can be applied to a startup in the following ways:

1. Improve the Product

- Customer feedback can be used to add, remove, or improve product features.

2. Improve the Value Proposition

- The startup can clearly communicate the main benefit that customers receive.

3. Select the Right Target Market

- Customer data helps identify the most suitable customer group.

4. Set Suitable Pricing

- Customer feedback can help the startup understand acceptable price levels.

5. Improve Customer Experience

- Identified pain points can be removed to make the product easier and more convenient to use.

6. Change the Business Idea if Necessary

- If the original idea does not solve an important problem, the startup can modify or **pivot** its idea.

Customer Discovery Flow

Identify Target Customers → Understand Problems → Conduct Research → Validate Problems → Analyse Data → Identify Patterns → Develop Solution → Test → Collect Feedback → Improve Startup

Conclusion

Customer Discovery is an important process for startups because it helps them understand customers before investing heavily in a product. By using interviews, surveys, observations, focus groups, and prototype testing, entrepreneurs can identify real problems and customer needs. Analysing these insights helps startups **develop better products, reduce risks, improve customer satisfaction, and increase the chances of business success**.

User Stories (US)

1. What is a User Story?

A **User Story (US)** is a short and simple description of a feature or requirement written from the **user's point of view**. It explains **who the user is, what the user wants to do, and what outcome the user expects**.

A common format of a user story is:

“As a [user], I want to [goal/action], so that [outcome].”

Example:

“As a student, I want to receive assignment reminders, so that I can submit my assignments on time.”

2. Components of a User Story

A. The User

- The **user** identifies the person who will use or benefit from the product or feature.
- It helps the development team understand **whose needs** are being addressed.
- The user can be a student, customer, employee, administrator, or any other target user.

Example:
In an online learning application, the user may be a **student**.

B. The Goal / Action

- The **goal or action** describes what the user wants to do with the product or service.
- It explains the specific function or requirement expected by the user.
- The action should be clear and focused on the user's need.

Example:
The student wants to **receive assignment reminders**.

C. The Outcome

- The **outcome** explains why the user wants to perform the action.
- It describes the benefit or result that the user expects.
- It helps the development team understand the real purpose behind the requirement.

Example:
The student wants reminders **so that assignments can be submitted on time**.

3. Complete User Story Example

Online Food Delivery System

User: College student
Goal/Action: Wants to pre-order food
Outcome: Wants to avoid waiting in long queues during breaks.

User Story:

"As a college student, I want to pre-order my food through the application, so that I can receive my food during my break without waiting in a long queue."

4. Benefits of Using User Stories in Design and Development

1. Focuses on User Needs

- User stories keep the development process focused on actual customer requirements.

2. Easy to Understand

- User stories are written in simple language and can be understood by both technical and non-technical team members.

3. Improves Communication

- They improve communication between customers, designers, developers, and other stakeholders.

4. Helps Prioritize Features

- User stories help teams decide which features are most important to users.

5. Supports Agile Development

- User stories are widely used in Agile development to organize and manage product requirements.

6. Improves Product Quality

- Understanding user expectations helps developers build more useful and relevant features.

7. Encourages Customer Involvement

- Customers can provide feedback and clarify their requirements during development.

8. Supports Iterative Development

- User stories can be modified and improved as new customer requirements are discovered.

5. Limitations of User Stories in Design and Development

1. Lack of Detailed Information

- A user story is short, so it may not contain all technical and functional details.

2. Can Be Ambiguous

- Different team members may interpret the same user story differently if it is not written clearly.

3. Difficult for Complex Requirements

- Very large or technically complex requirements may be difficult to describe using a single user story.

4. Depends on Customer Input

- Poor or incomplete customer feedback can result in incomplete or incorrect user stories.

5. May Ignore Technical Requirements

- User stories mainly focus on user needs and may not fully describe technical, security, or performance requirements.

6. Requires Continuous Updating

- User stories may need to be changed when customer needs or project requirements change.

7. Can Become Too Numerous

- A large project may contain many user stories, making them difficult to organize and manage.

6. User Story Process

Identify User → Understand Need → Define Goal/Action → Identify Expected Outcome → Write User Story → Discuss with Team → Develop → Test → Collect Feedback → Improve

Conclusion

User Stories are an effective technique for representing customer requirements from the user's point of view. They clearly describe the **user, goal/action, and expected outcome**. They improve communication, support Agile development, and help teams create customer-focused products. However, user stories may lack technical details and can become difficult to manage for complex projects.

Minimum Usable Prototype (MUP)

1. Concept of Minimum Usable Prototype (MUP)

A **Minimum Usable Prototype (MUP)** is the simplest version of a product or solution that contains enough important features to be **usable by real users** and to collect meaningful feedback.

- MUP focuses on creating a **small but usable product** rather than developing a complete product at the beginning.
- It allows designers and developers to test whether the basic solution works for users.
- Customer feedback from the MUP is used to improve the product.

- It helps reduce development time, cost, and risk.
- MUP is especially useful during the early stages of innovation and product development.

Example:

For a **food delivery application**, an MUP may include only restaurant selection, food ordering, basic payment, and order status tracking.

2. Key Elements of MUP

1. Core Function

- The MUP should contain the most important function needed to solve the user's main problem.

2. Usability

- The prototype should be simple enough for real users to understand and use.

3. Target Users

- The MUP should be designed for a clearly identified group of users.

4. Basic User Experience

- The interface should provide a clear and convenient experience without unnecessary features.

5. Feedback Mechanism

- The MUP should provide a way to collect feedback from users.

6. Testable Solution

- The prototype should be functional enough to test the main idea in a real or realistic situation.

3. Benefits of MUP

1. Reduces Development Cost

- MUP requires fewer resources because only essential features are developed initially.

2. Saves Time

- A simple prototype can be developed and tested faster than a complete product.

3. Collects Real User Feedback

- Real users can interact with the prototype and provide useful feedback.

4. Reduces Risk

- Problems can be identified before large amounts of money and effort are invested.

5. Improves Product Design

- User feedback helps developers improve the product according to actual needs.

6. Validates the Idea

- MUP helps determine whether the basic product idea is useful and practical.

4. Examples of MUP

Example 1: Food Delivery App

- The initial version may allow users to select a restaurant, order food, make payment, and track the order.

Example 2: Online Learning Platform

- The initial prototype may provide login, access to basic lessons, and simple quizzes.

Example 3: Online Shopping App

- The MUP may provide product search, product details, cart, and basic checkout.

Example 4: Cab Booking App

- The initial version may allow users to enter a destination, request a ride, and view basic driver information.

5. Other Minimum Products

There are several related minimum-product concepts used in product development.

1. MVP – Minimum Viable Product

- An **MVP** is the simplest product that provides enough value to early users and helps validate the business idea.

2. MLP – Minimum Lovable Product

- An **MLP** is the smallest product that not only works but also provides a **pleasant and attractive experience** that users enjoy.

3. MUP – Minimum Usable Prototype

- An **MUP** is the simplest usable prototype designed mainly to test functionality and usability with users.

6. Difference Between MLP, MVP and MUP

Feature	**MUP**	**MVP**	**MLP**
Full Form	Minimum Usable Prototype	Minimum Viable Product	Minimum Lovable Product
Main Focus	Usability and testing	Viability and value	User satisfaction and emotional appeal
Purpose	Test the basic solution	Validate the business/product idea	Create a product users like and want to continue using
Development Stage	Early prototype stage	Early product/market stage	More refined product stage
User Feedback	Mainly usability feedback	Product and market feedback	Experience and emotional feedback
Features	Only essential features	Minimum features needed to provide value	Minimum features with a strong user experience

7. Challenges of MUP

1. Limited Features

- An MUP contains only essential features, so users may feel that the product is incomplete.

2. Poor User Experience

- A very simple prototype may not provide a polished or attractive experience.

3. Misinterpretation of Feedback

- Users may judge the entire idea negatively because of limitations in the early prototype.

4. Technical Limitations

- Some features may be difficult to implement in the early version.

5. Difficulty in Selecting Features

- It can be challenging to decide which features are essential and which can be postponed.

6. Limited User Adoption

- Users may not be interested in using a prototype that has very few features.

8. Strategies to Overcome MUP Challenges

1. Clearly Identify the Core Problem

- The development team should clearly understand the main customer problem before selecting features.

2. Prioritize Essential Features

- Only features that directly solve the main problem should be included in the first version.

3. Conduct User Testing

- The MUP should be tested with a suitable group of target users.

4. Collect Continuous Feedback

- Feedback should be collected regularly and used to improve the prototype.

5. Explain the Prototype Clearly

- Users should understand that the MUP is an early version and not the final product.

6. Improve Iteratively

- The prototype should be continuously improved based on user feedback and test results.

MUP Development Process

Identify User Problem → Select Essential Features → Develop MUP → Test with Users → Collect Feedback → Analyse Results → Improve Prototype → Develop Better Product

Conclusion

A **Minimum Usable Prototype (MUP)** helps designers and entrepreneurs create a simple but usable version of a product for early testing. It reduces **time, cost, and development risk** while providing valuable user feedback. By carefully selecting essential features, conducting user testing, and continuously improving the prototype, organizations can develop products that better satisfy real customer needs.
